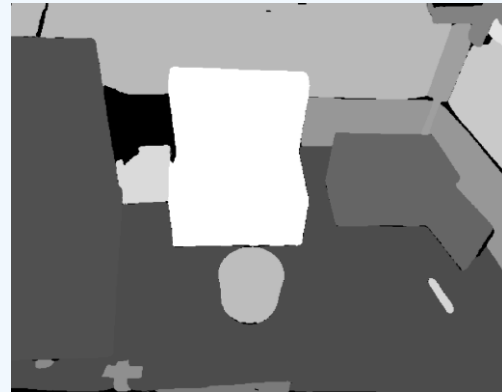


Object-level physical proxy

Calibrated RGB + instance masks

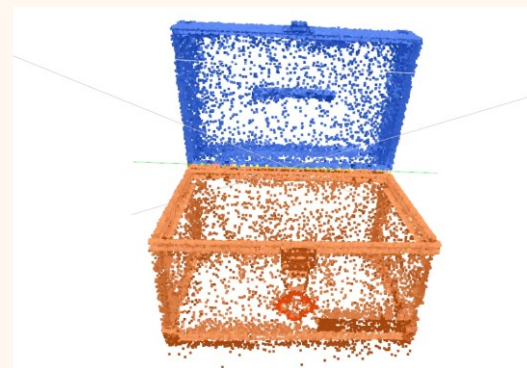


RGB (calibrated)



Instance masks

Object-level 3DGS



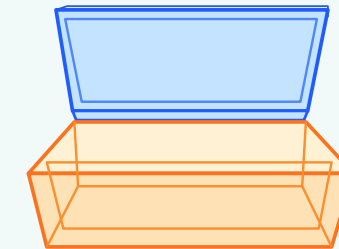
Gaussian object (3DGS)

Visual asset



Appearance (from 3DGS)

Simplified collision asset



Lid/base collision model

Isaac Sim proxy



Measurements / CAD / URDF / priors



Recorded physics

Joints · Mass · Contact

Current depth



Scale + alignment

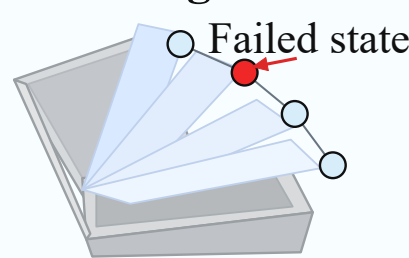
Failure-region expansion

Current-policy rollout



Execute current VLA policy in Isaac Sim

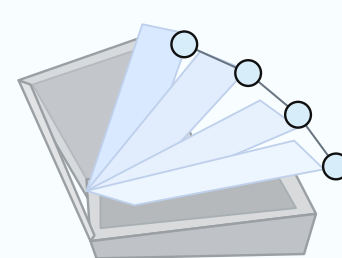
Failure neighborhood



Local structural state space

● = failed state
○ = nearby state

Feasible initial states



Local structural state space (continuation)
Sample feasible initial states near failure

Candidate rollouts

Current VLA from sampled states

$s_0^{(1)}$	a_1	a_2	$\cdots$	a_T
$s_0^{(2)}$	a_1	a_2	$\cdots$	a_T
$\vdots$				
$s_0^{(3)}$	a_1	a_2	$\cdots$	a_T

Optional residual RL (simulation only)

Joint acceptance

Task success ☒

Physics ☒

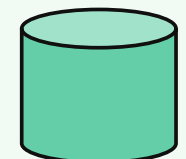
Safety ☒

Any failed

☒ Discard

All pass

Successful complete trajectories



SUCCESSFUL COMPLETE trajectories

Updated policy



Real demonstrations

Real-dominant retraining